Calculate yeer Waterline!

STEP 1: CALCULATE DISPLACEMENT VOLUME

Using Archimedes' principle, we determine the volume of water displaced that equals the boat's total weight.

$$WaterDensity = 1g/cm^3$$

$$DisplacementVolume\,(cm^3) = \frac{Weight\,(g)}{WaterDensity\,(g)} = \frac{Weight\,(g)}{1} = Weight\,(g)$$

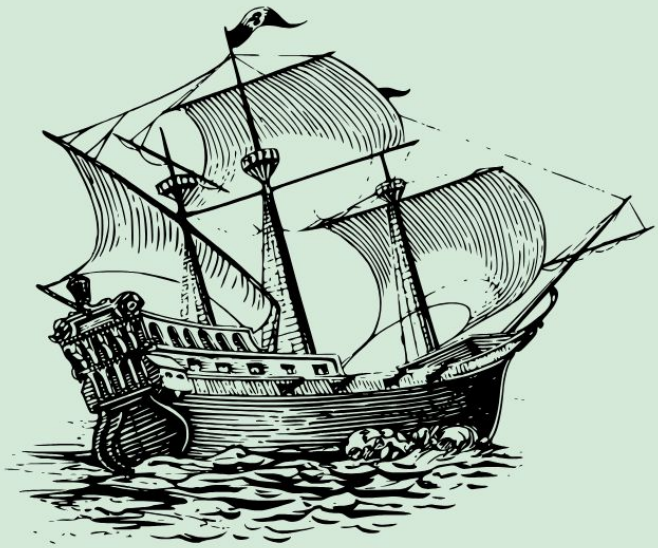
STEP 2: CALCULATE THE WATERLINE AREA

For a flat-bottomed boat, this is just the area of the bottom (length x width for a rectangular boat).

STEP 3: CALCULATE THE WATERLINE HEIGHT

To find the waterline height, divide the displacement volume by the waterline area.

$$WaterlineHeight = \frac{DisplacementVolume\,(step1)}{WaterlineArea\,(step2)}$$



Calculate yeer Waterline!

STEP 1: CALCULATE DISPLACEMENT VOLUME

Using Archimedes' principle, we determine the volume of water displaced that equals the boat's total weight.

$$WaterDensity = 62.4 lbs/ft^3 = 0.0361 lbs/in^3$$

$$DisplacementVolume = \frac{TotalWeight}{WaterDensity}$$

STEP 2: CALCULATE THE WATERLINE AREA

For a flat-bottomed boat, this is just the area of the bottom (length x width for a square boat).

STEP 3: CALCULATE THE WATERLINE HEIGHT

To find the waterline height, divide the displacement volume by the waterline area.

$$WaterlineHeight = \frac{DisplacementVolume\,(step1)}{WaterlineArea\,(step2)}$$

